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Web and Social Network Analytics

Retail Analytics and Web Traffic Analysis

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Part 1: Retail Analytics Report - HighlandGear

1. Theoretical Framework

The optimization process for HighlandGear's marketing and loyalty strategies. I applied to Association Rule Mining and Social Network Analysis (SNA).

The Association Rule Mining technique reveals hidden patterns which show how different products relate to each other. The analysis needs Lift as its primary performance metric.

Social Network Analysis (SNA) models customer referrals by using directed graph structures. The Degree Centrality method identifies "Power Referrers" through their total number of referrals.

The Clustering Coefficient tracks triadic closures. The network shows high clustering which indicates a "Small World" pattern, but HighlandGear's network organization presents two distinct problems which result in brand diffusion obstacles through its fragmented structure and Echo Chambers that create separate isolated areas.

2. Data Analysis & Insights

The evaluation of *transactions.csv* and *referrals.csv* produced these evidence-based findings:

Product Bundling & Associations: The "Sleeping Bag and Tent" combination appears 758 times because customers fall into the "Entry-Level Camper" segment.
The "Camping Stove and Dry Bag" combination produced the highest Lift value which reached 4.75. This indicates a specialized persona where the purchase of one item makes the other nearly five times more likely a high-value target for cross-selling.

Influencer and Network Structure: The current referral network shows no clear organization because its density measures at 0.000758, which indicates it exists at an initial development phase.

Top Referrers: Customers C00020, C00152, and C00036 share the highest Degree Centrality (0.0215). They function as the main systems which drive the process of acquisition.

The "Zero" Clustering Insight: The Average Clustering Coefficient is 0.0000 confirms no triadic closure. While this prevents echo chambers. The company struggles to keep customers because new customers only interact with their first referral.

3. Strategic Recommendations

The "Basecamp Starter Kit": Formalize a bundle of the Tent, Sleeping Bag and Camping Mat. The "high-support" bundle focuses on the most popular customer segment because this group shows the greatest mutual trust which leads to the highest possible Average Order Value (AOV).

The program needs to offer exclusive rewards to its top referrers who achieve C00020 status. Since our data shows these influencers are also technical buyers, "Early Access" to new gear will incentivize them to bridge their isolated referral chains.

Strategic Interconnection: HighlandGear needs to create referral incentives for "Bridge" connections which provide benefits to customers who bring new users from different social networks to expand their network reach.

4. Ethics & Regulatory Considerations

Our system tracks all referral link processing activities which lead to social profiling activities that must follow GDPR regulations. HighlandGear needs to implement "Privacy by Design" which requires data collection only after users provide direct consent for data collection and the company must follow the "Right to be Informed" requirement.
The use of Degree Centrality as a single fairness metric for algorithms does not identify "Quiet Loyalists" who spend heavily but have minimal social connections. A fair system needs to merge purchase frequency data with social influence metrics to stop discriminatory practices from happening.
The EU AI Act together with transparency requirements establish a Right to Explanation which enables customers to understand the automated process which creates their bundles. HighlandGear needs to show all customer data usage activities because this approach will help the company defend its customer trust.

5. Critical Reflection

The research provides an effective framework for expansion, but the study presents a single moment in time. A Gold Standard research method requires researchers to conduct a long-term study which follows influencer network centrality changes throughout time while controlling annual changes in camping equipment market trends.

Part 2: Web Analytics Report - HaggisBus

1. Theoretical Framework

The evaluation of HaggisBus booking journey digital effectiveness required me to use four essential Web Analytics metrics.

The Conversion Rate (CR): indicates the percentage of distinct website visitors who finished buying their products through the *purchase_success* event. The metric shows website operational efficiency because it produces the highest possible revenue output.

The Bounce Rate: metric shows the percentage of users who leave the platform after visiting only one page (Session Depth =1) suggest a mismatch between marketing creative and page content.

The Session Depth: The average number of pages viewed per session as a primary indicator of brand engagement.

The marketing funnel analysis: monitors user behaviour from Home to Pricing and then to Purchase to identify when customers abandon their journey because of technical or psychological barriers.

2. Campaign & Channel Performance

The analysis of more than 160,000+ sessions shows that different channels generate significantly different Return on Ad Spend (ROAS) results.

The High-Performers: Direct traffic leads the portfolio with a 31.67% Conversion Rate which is followed by *linkedin_advert* at 25.20%. The Bounce Rate for LinkedIn and Partner adverts maintained a 0.00% rate which shows these channels attract leads who have already demonstrated their interest in the product.

Facebook Performance Gap: The *facebook_advert* platform operates as the source which creates current financial waste. It exhibits a massive 42.62% Bounce Rate and an extremely poor conversion rate of just 0.93%. Facebook uses a present-day target strategy which aims to acquire new users who do not want to purchase because they cannot value the website.

3. User Experience & Platform Analysis

The data shows that users face a major "Mobile Gap" which creates a significant obstacle during their last step of booking.

Platform Disparity: The conversion rates for Windows desktop users (16.40%) and Mac desktop users (16.00%) exceed those of mobile users. The conversion rate for iOS users amounts to 12.69% which shows a 4% lower performance than what desktop users obtain. The situation demands an immediate assessment of how mobile Safari browser functions.

The "Final Friction" Leak: Our funnel analysis reveals that HaggisBus keeps 81.8% of users who move from the Landing Page to the Pricing Page, but the user experience ends at this point. The "Address Entry" stage shows that 54.6% of users who make it to this point will abandon their purchase. The final form demonstrates a 45.4% decrease in customer base which shows that customers struggle with checkout fields and encounter unexpected costs when they want to make a purchase at their most active time. Blog 2 (Inspirational content) achieved success through its 17.64% Conversion Rate. The performance of Blog 1 (Instructional) stands at 9.96% which indicates worse results than users who did not visit any blog content. The blog seems to divert users away from their purchase journey.

4. Ethics & Strategic Recommendations

Budget Optimization: The company needs to remove all Facebook advertising expenses because it should use its advertising budget to support LinkedIn advertising and Blog 2 marketing efforts. The data shows these channels serve as the best methods for acquiring new customers. The 45% "*Address-to-Success*" leak at HaggisBus needs "*One-Click Checkout*" with Apple Pay or Google Pay to resolve mobile form-filling problems. The system requires Clickstream tracking to function because it needs to analyse complete behavioural data. The organization needs to adopt "*Privacy by Design*" principles which demand that they get users to approve tracking cookies directly and they must stop using "*Unknown*" platform identifiers which help tracking systems bypassing user privacy settings. The system must prevent discriminatory effects which could result from dynamic pricing and promotional codes that affect users based on their device platform between iOS and Android while following the EU AI Act requirements for price disclosure.

5. Critical Reflection

The quantitative analysis shows which areas of revenue loss occur, but it fails to provide the underlying reasons for these losses. The data shows behavior at a single point in time. The "*Gold Standard*" of business intelligence at HaggisBus requires combining these results with qualitative UX testing which includes heatmaps and user interviews to determine why users abandon the address entry process at such a high rate.